



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to VANMATHI AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

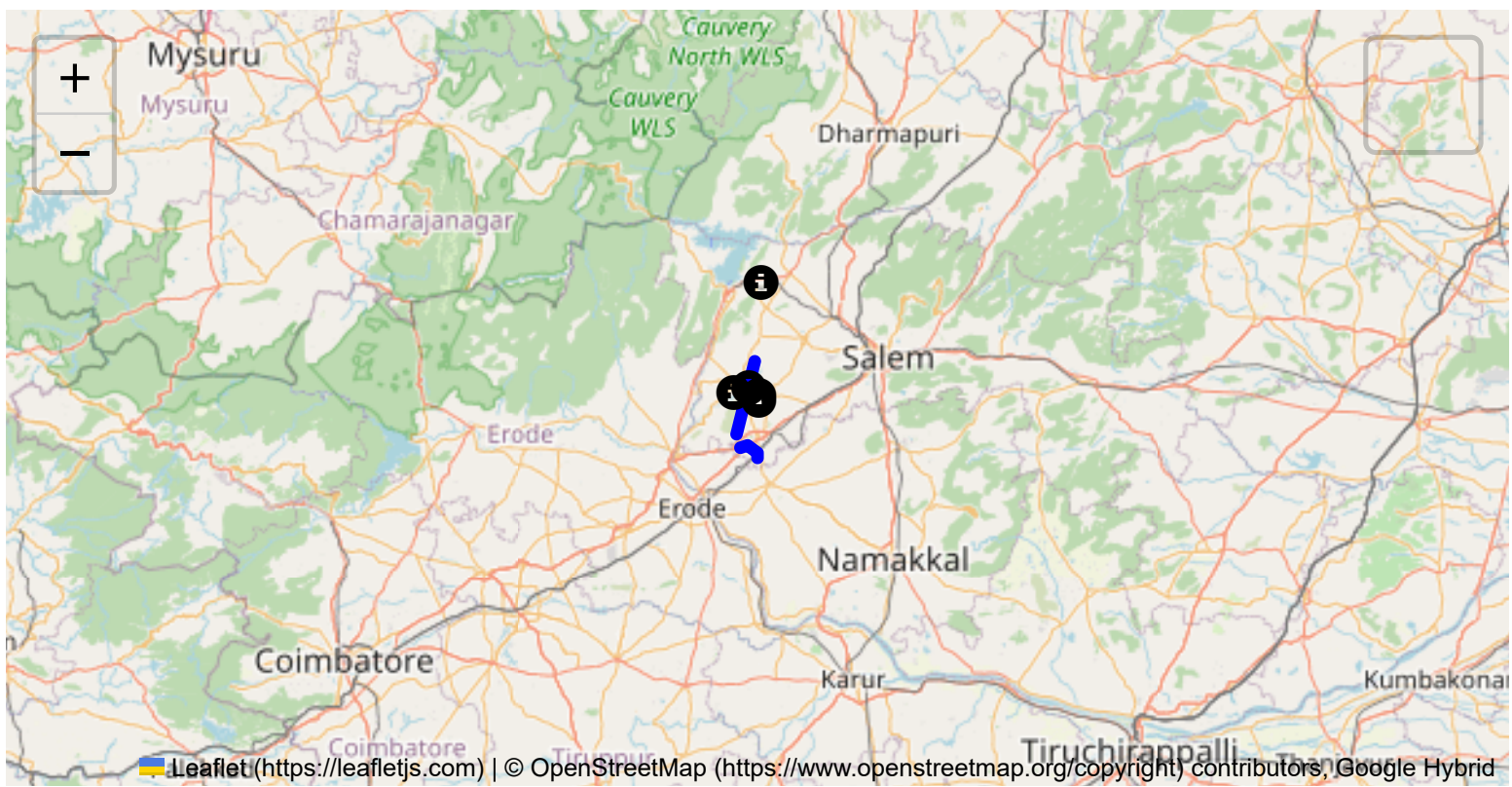
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

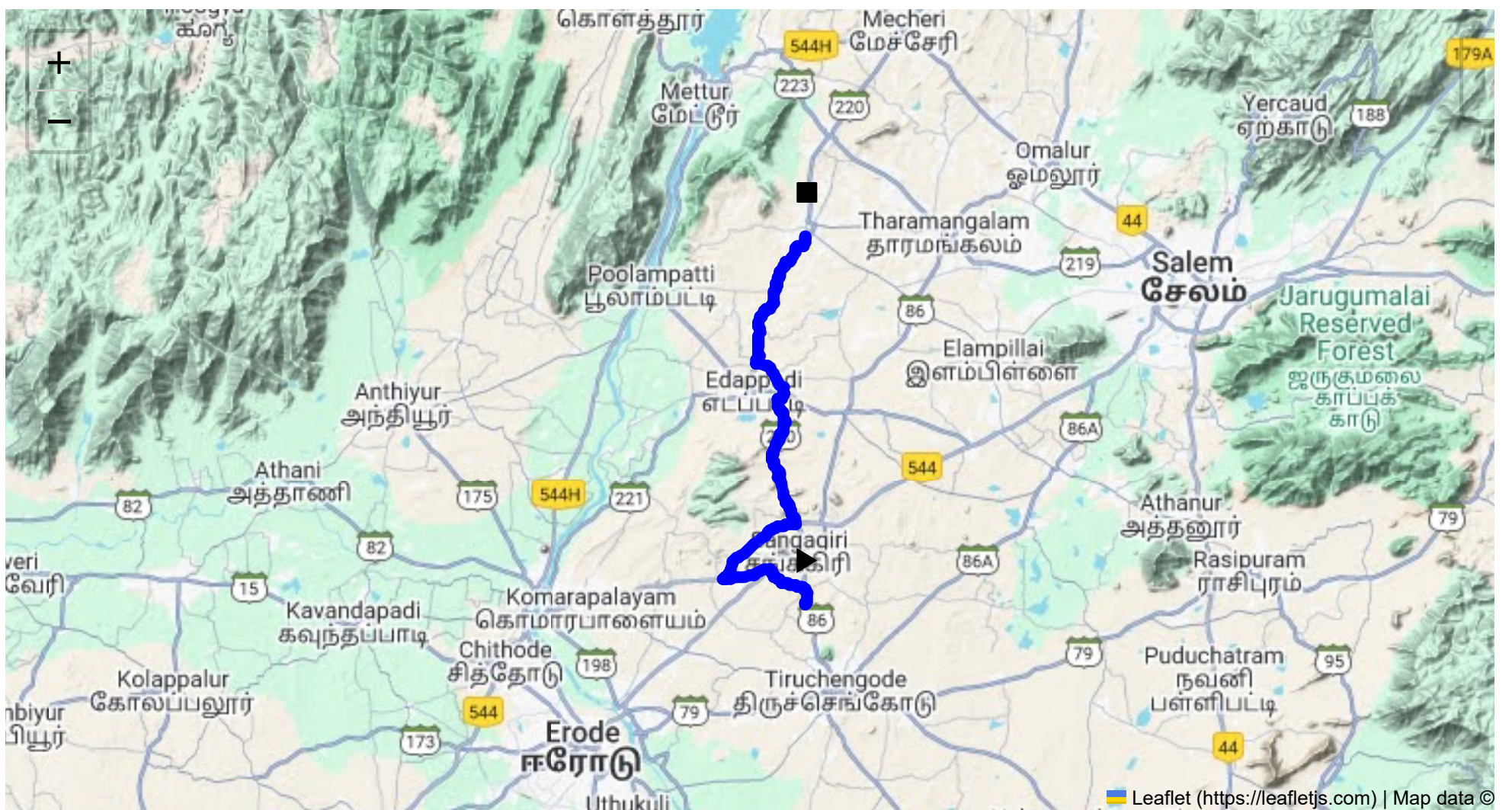
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 41.81 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.691791, 77.874047)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route from Sangagiri to Jalakandapuram in Tamil Nadu covers approximately 41.81 kilometers. The journey begins at CVQF+23W in Sangagiri and proceeds northwest on local roads, connecting to key points via Sankari RS - Erode Privu Road, Bhavani Main Road through Sankari West, and Padaiveedu before reaching the destination at MVRF+MMJ, SH 220 in Jalakandapuram.

2. Typical Weather Conditions and Potential Weather-Related Hazards: Tamil Nadu experiences tropical weather with high temperatures and humidity, particularly in summer. Monsoon seasons (June-September) may bring heavy rains that can cause poor visibility and flooding on roads, leading to potential delays and increased accident risk. During the dry months, dust can also impair visibility.

3. Analysis of Traffic Patterns: Traffic congestion is often significant during peak hours, approximately 8:00-10:00 AM and 5:00-8:00 PM, especially near urban centers like Sangagiri and at intersection points on Sankari RS - Erode Privu Road and Bhavani Main Road. The area around Sankari usually sees heightened traffic due to commercial activities and local transportation.

4. Assessment of Road Quality and Infrastructure: The route comprises a mix of state highways and local roads. Overall, road conditions are fair, but expect variability with occasional potholes, narrow segments, and limited lighting. Infrastructure such as signaling and signage might not be consistent, particularly on less-maintained segments.

5. Suggestions for Alternative Routes for Emergencies: In emergencies, consider bypassing local roads by using NH544 for faster evacuation and better-managed road conditions. However, note that using alternative state highways might involve longer travel distances.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Tamil Nadu has strict regulations for the transport of hazardous materials, with requirements for permits and specific routing to minimize risks. Registration with the local transportation authority and compliance with safety equipment and protocols are mandatory.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials: Some incidents with heavy vehicles reported in the past involve overturns and spills, primarily due to speeding or inadequate road safety measures. The absence of timely information on hazardous material incidents suggests these are less frequent but still a concern.

8. Environmental Considerations and Sensitive Areas: The route passes through semi-urbanized and rural landscapes. Potential environmental concerns include nearby agricultural fields where accidental spills could affect crop safety. Rural areas may also have less enforcement of environmental regulations.

9. Analysis of Communication Coverage: Mobile network coverage is generally good along major roads and urban centers. However, there could be dead zones in rural segments or less populated areas where network providers have sparse infrastructure.

10. Estimated Emergency Response Times for Different Route Segments: Emergency response times vary along the route. Proximity to urban centers like Sangagiri usually allows for a faster response, approximately 15-30 minutes. Remote or rural areas may experience delays ranging from 30 minutes to over an hour.

12. Overall Summary of Risk Assessment: The route presents moderate risk levels for transporting hazardous materials, primarily due to weather variability, infrastructure inconsistencies, and traffic congestion. Risks can be mitigated with proper planning, awareness of local traffic patterns, and emergency preparedness. Emphasizing compliance with safety regulations and best practices in driving can significantly reduce potential hazards. Additionally, constant communication and using technology for real-time updates can help anticipate and respond to any issues encountered during transport.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.45357, 77.85789	30 KM/Hr
6	Turn	Medium	11.45352, 77.85698	30 KM/Hr
7	Turn	Medium	11.46048, 77.85036	30 KM/Hr
8	Turn	Medium	11.46303, 77.84945	30 KM/Hr
9	Turn	Medium	11.46318, 77.84913	30 KM/Hr
10	Turn	High	11.45542, 77.81725	15 KM/Hr
11	Turn	High	11.45634, 77.81673	15 KM/Hr
12	Blind Spot	Blind Spot	11.49390, 77.86748	10 KM/Hr
13	Turn	Medium	11.49404, 77.86724	30 KM/Hr
14	Turn	Medium	11.50866, 77.86053	30 KM/Hr
15	Turn	Medium	11.55039, 77.85511	30 KM/Hr
16	Turn	Medium	11.55460, 77.85624	30 KM/Hr
17	Turn	High	11.57903, 77.85432	15 KM/Hr
18	Turn	Medium	11.60727, 77.84032	30 KM/Hr
19	Turn	Medium	11.60972, 77.84164	30 KM/Hr
20	Turn	Medium	11.64064, 77.84611	30 KM/Hr
21	Turn	Medium	11.67373, 77.86230	30 KM/Hr
22	Turn	Medium	11.68171, 77.86676	30 KM/Hr
23	Turn	High	11.68490, 77.86893	15 KM/Hr
24	Turn	High	11.68472, 77.87281	15 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58300, 77.85813



Risk Type: Roundabout
Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.60348, 77.83967



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46048, 77.85036



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46318, 77.84913



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45634, 77.81673



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49390, 77.86748



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49404, 77.86724



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50866, 77.86053



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55039, 77.85511



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55460, 77.85624



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57903, 77.85432



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60727, 77.84032



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60972, 77.84164



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64064, 77.84611



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67373, 77.86230



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68171, 77.86676



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.68490, 77.86893



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.68472, 77.87281
